THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 04

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** dark pink or magenta	natural indicator
2	**(d)** Lemon juice	citric acid gives the sour taste
3	**(b)** conductors	metals
4	**(d)** can conduct electricity	safety
5	**(a)** Graphite	form of carbon
6	**(c)** ductility	e.g. copper wire
7	**(d)** iron oxide	reddish-brown
8	**(b)** combustible	e.g. wood, LPG
9	**(c)** 28–30 days	may vary 21–35
10	**(c)** 3 to 7 days	bleeding phase
11	**(d)** condensation	cooling
12	**(c)** reflect more heat	keep us cooler
13	**(b)** metre per second	m/s
14	**(c)** odometer	total distance
15	**(b)** breathing	physical process
16	**(d)** fats	breaks fats into droplets
17	**(b)** transpiration	pull from leaves
18	**(d)** sunlight	energy source
19	**(b)** splitting of white light	dispersion
20	**(d)** decreasing	towards new moon

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	To neutralise it and protect water bodies (2).	2
22	It is a good conductor and ductile (2).	2
23	Metal oxides basic (1); non-metal oxides acidic (1).	1+1
24	Harassing others with digital devices (1); inform parents/teachers (1).	1+1
25	Metal conducts heat away from the body (2).	2
26	Sundial; water clock (2).	2
27	Moves downward and ribs move up and out, so air enters (2).	2
28	Cannot photosynthesise and make food (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Circle with cross, several cell symbols, break with lever (3).	3
30	Digestion; respiration (3).	3
31	Warmth from the Sun, heat from a bonfire, microwave (3).	3
32	Neutralises acid, breaks fats (3).	3
33	Smooth water reflects regularly (1.5); ripples scatter light (1.5).	1½+1½
34	Opposite hemispheres tilt towards or away from Sun (3).	3
35	Lemon has citric acid (1.5); soap is a base (1.5).	1½+1½
	OR: Shows the point where the solution becomes neutral (3).	3

36	No gas exchange or transpiration (1.5); photosynthesis stops (1.5).	1½+1½
	OR: CO_2 enters and O_2 leaves through stomata (1.5); reverse in respiration (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Heat passes along the metal (2). (ii) Rod with wax and pins (2). (iii) Plastic is a poor conductor (1).	2+2+1
	OR: (i) Table with observations (3). (ii) Classification (2).	3+2
38	(i) Needs nutrients for growth (1). (ii) Iron, protein (1). (iii) Eat balanced meals (1). (iv) Doctor/dietitian (2).	1+1+1+2
	OR: (i) Desire to belong (1). (ii) Positive habits or harmful ones (2). (iii) Think, say no, seek guidance (2).	1+2+2
39	(i) Using Earth's rotation and revolution (2). (ii) One rotation (1). (iii) One revolution around the Sun (2).	2+1+2
	OR: (i) Travel planning, sports (2). (ii) Example computation (2). (iii) Consistency (1).	2+2+1
40	(i) Straight, reflected, forms shadows (2). (ii) Card-hole and mirror activity (2). (iii) Vision, energy (1).	2+2+1
	OR: (i) Dark patch vs picture (2). (ii) Erect virtual vs inverted real (2). (iii) Pinhole image (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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